[225Ac]Ac-AKY-1189 (Aktis Oncology)

anti-Nectin-4 alpha-particle radiopharmaceutical — actinium-225 miniprotein radioconjugate for locally advanced or metastatic urothelial cancer (lead)

June 2026

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Alpha-RLT on validated Nectin-4 target; human efficacy read not before Q1 2027

Phase 1b: NECTINIUM-2 (NCT07020117) · Aug 2025

Asset

– [225Ac]Ac-AKY-1189 · Ac-225 miniprotein radioconjugate
– Nectin-4 target
– [64Cu] PET companion (CDx unvalidated)
– IV up to 6 cycles

Value & Feasibility

– Post-EV/pembro mUC unmet need
– Target de-risked by Padcev (EV-302 OS HR 0.47)
– Alpha-particle kill bypasses MMAE-resistance pathways
– PET gate converts expression to quantitative patient selection

Feasibility

– No human anti-tumor efficacy
– 3 PCT families pending; no granted patents
– Ac-225 supply uncontracted
– FDA Oct 2025 dosimetry guidance may require protocol amendment

Risk

– EV-202 on-target failures in TNBC/NSCLC weaken expansion basket
– Zelenectide + LY4052031 3–4 yr ahead; step-edit gating compresses penetration
– Lilly ~10% equity + competing LY4052031 ADC

Valuation: ~$1.7B market cap (Jun 2026); control bid >$2.2B exceeds pre-PoC rNPV; Q1 2027 Part 1 read is the valuation-adjudicating catalyst

Sources: NECTINIUM-2 (NCT07020117) • Padcev EV-302 data and commercial update — pharmaphorum

Same Nectin-4 target, a different weapon: the approved antibody smuggles chemo into one cell at a time, while [225Ac]Ac-AKY-1189 kills with alpha-radiation from the cell surface and reaches neighbors too

940–1,690 peak mUC patients at ~$170K net; 10% POS anchors the rNPV

Post-EV la/mUC is the primary value driver; expansion basket carries elevated execution risk after EV-202 on-target failures in TNBC/NSCLC.

Take-home: At 10% POS and base penetration, rNPV sits below the >$2.2B control premium implied by Aktis's current market cap — the market is pricing in higher POS or broader basket penetration than our model; Q1 2027 Part 1 safety/dosimetry data adjudicates both.

Funnel step	US pts/yr
la/mUC incident	~13,000
Receive systemic therapy	~9,750
Reach 2L+, remain fit	~4,700
Nectin-4+ / PET-eligible	~3,750
Peak treated — base / bull	940 / 1,690

Sources: ACS Cancer Facts & Figures 2024 • Pluvicto $42,500/dose WAC — Novartis press release • BIO Clinical Development Success Rates 2011–2020 • Padcev + Keytruda cost analysis — Prime Therapeutics

Abbreviations used in this assessment

Valuation & IP

POS probability of success
PTRS probability of technical & regulatory success

LOA likelihood of approval
rNPV risk-adjusted net present value

NPV net present value
DCF discounted cash flow
EV enterprise value
WAC wholesale acquisition cost
GtN gross-to-net

Disease & clinical

OS overall survival
PFS progression-free survival
ORR objective response rate

Regulatory & trials

FDA US Food and Drug Administration

NDA New Drug Application

Mechanism & nonclinical

ADC antibody-drug conjugate
IV intravenous
SC subcutaneous

$170k/patient net price aligns with established RLT and Nectin-4 ADC comps

Net price modeled at ~$170k: $47k/dose WAC × 4.5 avg doses × 80% GTN, calibrated against approved RLT courses and same-target ADC.

340B drag: academic RLT centers disproportionately 340B; blended GTN modeled at 80%, rising to ~85% post pass-through.

Take-home: $170k net sits 25–35% below Pluvicto/Padcev — conservative for a novel actinium RLT; 340B at academic centers is the primary GTN risk.

Analog	Indication	Annualized WAC	Modality
Pluvicto ([^{177}Lu]PSMA-617)	mCRPC PSMA+	$255,000/course	RLT beta-emitter
Lutathera ([^{177}Lu]DOTATATE)	GEP-NETs	$220,000–$244,000/course	RLT beta-emitter
Xofigo (Ra-223)	mCRPC bone-mets	~$185,000/course [est.]	RLT alpha-emitter
Padcev (enfortumab vedotin)	Metastatic UC, Nectin-4+	~$230,000/yr	ADC, same target

Sources: Pluvicto FDA approval — $42,500/dose WAC confirmed (Novartis, Mar 2022) • Padcev + Keytruda cost analysis — $189k–$204k net annual (Prime Therapeutics, 2024) • Lutathera buy-and-bill pricing, HCPCS A9513 — $55,000–$61,005/dose • Xofigo FDA prescribing information, NDA 203971

PTRS + BIO/Citeline LOA converge on 8.4% point POS (range 5–15%)

POS built bottom-up: Gosset PTRS anchors the Ph1→Ph2 transition; BIO/Citeline oncology LOA — adjusted for validated target, Fast Track, and accelerated-approval plausibility — sets the remaining-to-approval leg.

Take-home: The 8.4% point POS sits within the all-oncology historical band; the critical swing is whether Part 1/2 ORR in post-EV mUC supports an accelerated-approval pathway without a randomized Phase 3.

Component	Value	Basis
Gosset PTRS Ph1→Ph2	38.2%	NCT07020117; positive SHAP: sponsor rate (+0.38), urothelial (+0.26), Nectin-4 (+0.15); negative: first-in-class penalty (−0.26), endpoint count (−0.24)
External Ph2→approval LOA	12–38%	BIO/Citeline oncology 2011–2020; Fast Track (Feb 2026) + validated-target uplift
Combined POS (point)	**8.4%**	0.382 × 0.22
Historical cross-check	5–10%	BIO/Citeline ~5.3%; Wong et al. ~7.6% all-oncology Ph1→approval

Sources: NCT07020117 — NECTINIUM-2 Phase 1b (Aktis Oncology), ClinicalTrials.gov • BIO / Citeline "Clinical Development Success Rates 2011–2020" (2021) • Wong et al. "Estimation of Clinical Trial Success Rates" — Biostatistics 2019

rNPV $63M–$273M; market price implies ~36% POS on the bull scenario

rNPV derived from global peak-revenue scenarios (US × 1.85 ex-US gross-up) discounted at 2.5× to approval and risk-adjusted at the 8.4% point POS. Bull scenario requires multi-tumor regulatory approvals beyond lead mUC.

Scenario	Treated (US)	Global peak (unadj.)	NPV @ approval (2.5×)	POS	rNPV
Base — urothelial 2L+	940	$300M	$750M	8%	**$63M**
Bull — multi-tumor basket	4,130	$1.30B	$3.25B	8%	**$273M**

Sources: NCT07020117 — NECTINIUM-2 Phase 1b, ClinicalTrials.gov • BIO / Citeline "Clinical Development Success Rates 2011–2020" (2021) • Novartis FY2024 results — Pluvicto $1.39B (+42%) commercial context

EV+pembrolizumab reset 1L mUC; no approved post-EV standard defines the gap

EV+pembrolizumab (EV-302: OS HR 0.47, PFS HR 0.45) is now the preferred first-line standard across EAU/NCCN/NICE, leaving a defined post-EV refractory gap with no guideline-endorsed successor.

Take-home: The post-EV, post-PD-1/L1 refractory setting has no approved standard — this is AKY-1189's primary target population in mUC.

Line	Regimen	Key Efficacy	Guideline Status
1L preferred	EV + pembrolizumab	OS HR 0.47; PFS HR 0.45; ORR 67.7% (EV-302)	EAU / NCCN / NICE TA1097
1L cisplatin-eligible	Gemcitabine + cisplatin	Induction backbone	All-guideline preferred
1L cisplatin-ineligible	Gemcitabine + carboplatin	Standard compromise	Guidelines-endorsed
Switch maintenance	Avelumab	OS HR 0.69 vs BSC (JAVELIN Bladder 100)	Non-progressing post-platinum
2L	Pembrolizumab	OS HR 0.73 vs chemo (KEYNOTE-045)	Preferred post-platinum
2L	EV monotherapy	OS HR 0.70; ORR 40.6% (EV-301)	After platinum + PD-1/L1
2L+ (FGFR2/3+)	Erdafitinib	OS HR 0.64 (THOR)	FGFR-altered ~15–20% only
Later	Sacituzumab govitecan	ORR ~27% (TROPHY-U-01)	Heavily pretreated

Sources: EV-302/KEYNOTE-A39 (NCT04223856) • JAVELIN Bladder 100 (NCT02603432) • KEYNOTE-045 (NCT02256436) • EV-301 (NCT03474107)

Alpha emission + miniprotein scaffold circumvent MMAE efflux and trafficking dependence

[225Ac]Ac-AKY-1189 couples ^{225}Ac alpha emission (four α-particles per decay chain, high LET ~100 keV/µm, ~50–100 µm path) with an engineered miniprotein scaffold for rapid tumor uptake and fast normal-organ clearance — mechanistically distinct from the MMAE ADC class and beta-emitter concepts.

Take-home: Alpha kill is cell-cycle– and oxygenation-independent; neither mechanism overlaps with MMAE's known post-EV resistance pathways — the therapeutic rationale for a post-EV salvage trial is mechanistically grounded.

Attribute	EV / Padcev (Nectin-4 ADC)	Beta-emitter RLT	AKY-1189 (alpha miniprotein)
Cytotoxic principle	MMAE microtubule poison	Beta radiation (low LET, mm-range)	Alpha radiation (high LET, ~50–100 µm)
Drug efflux resistance	Susceptible — MMAE is P-gp substrate	Independent	Independent
Trafficking dependence	High — internalization, cleavage, release required	Moderate	Low — energy deposited on decay
Cross-fire range	Limited bystander	Long-range (normal-tissue dose risk)	Short-range — reaches antigen-low neighbors
Normal-organ clearance	IgG, multi-day circulation	Antibody/peptide dependent	Small binder, rapid wash-out
Patient selection	Assumed pan-Nectin-4	Variable	[64Cu] PET-confirmed positive only

Sources: NECTINIUM-2 (NCT07020117) • Pluvicto FDA approval — alpha-RLT commercial precedent (Novartis, Mar 2022) • Xofigo FDA prescribing information — alpha-emitter precedent (accessdata.fda.gov)

PET imaging confirms deep tumor uptake; alpha-RLT precedent is commercially validated

EORTC 2024 imaging showed strong AKY-1189 uptake in mUC, TNBC, NSCLC, and cervical tumors with limited normal-tissue signal; approved RLT agents (Xofigo, Pluvicto, Lutathera) exceeded $2B combined FY2024 revenue.

Target prevalence: Nectin-4 positive in ~80–90% UC, ~58–60% TNBC, ~54% NSCLC, ~83–86% cervical/HNSCC

Human imaging (EORTC 2024): Substantial tumor uptake across five histologies, limited normal-tissue exposure — supports wide therapeutic window

FDA Fast Track granted Feb 24, 2026 for mUC post-prior-systemic therapy

NECTINIUM-2 (NCT07020117): Phase 1b, n≈150, 9 US sites incl. MD Anderson & City of Hope; started Aug 2025; Part 1 safety/dosimetry readout Q1 2027

Modality precedent: Xofigo, Pluvicto, and Lutathera all approved; combined Novartis RPT revenue ~$2.1B FY2024

Take-home: Biomarker-gated design and established RLT infrastructure reduce key execution unknowns — no efficacy data yet; Q1 2027 Part 1 readout is the first value-inflection gate.

Sources: NECTINIUM-2 (NCT07020117) • Novartis FY2024 results — Pluvicto $1.39B; Lutathera ~$720M • Pluvicto FDA approval (Novartis, Mar 2022) • Lutathera FDA prescribing information (accessdata.fda.gov)

Antigen loss, daughter recoil, and ERA-223 define three independent risk layers

	Assessment	Observations	Rating
Nectin-4 Biology	**Antigen loss in post-EV disease**	– ~59% of metastatic lesions lose Nectin-4 vs primary tumor, enriching the post-EV population AKY-1189 targets for low-expression disease	⚪⚪🔴
	Efflux-resistant tumor biology	– EV-resistant models upregulate ABC transporters, and alpha radiation bypasses efflux but these tumors remain biologically aggressive after EV	⚪🟠⚪
Alpha-Emitter Precedent	**ERA-223 (Ra-223 + abiraterone, mCRPC)**	– Unblinded early with fractures 29% vs 11% and OS numerically worse (30.7 vs 33.3 mo, HR 1.195), showing cumulative alpha dose can flip net benefit	⚪⚪🔴
On-Target and Dosimetric Toxicity	**Nectin-4 skin expression**	– Nectin-4 is a normal adhesion molecule in skin epithelia, with EV driving rash in ~50% and a radioligand adding alpha-mediated radiation dermatitis risk	⚪🟠⚪
	Ac-225 daughter recoil	– Fr-221 and Bi-213 redistribute to kidney (~40%) and marrow after chelator ejection, with rapid tumor internalization the intended mitigation but untested at therapeutic dose	⚪⚪🔴

🟢 No major issues 🟠 Potential risks, addressable 🔴 Major issues / no mitigation

Sources: [ERA-223 trial — Smith et al., Lancet Oncol 2019]() • [Padcev (enfortumab vedotin) prescribing information (DailyMed)]() • [NCT07020117 — NECTINIUM-2 Phase 1b (Aktis Oncology)]()

NECTINIUM-2 Evidence

15/15 patients show tumor uptake; human anti-tumor efficacy is undemonstrated

	Assessment	Observations	Rating
Preclinical (Company-Reported, Single Cell Line)	**Xenograft efficacy, high-Nectin-4**	– 53% mean tumor regression at Day 25 and 75% mouse survival at 8 wk vs 0% vehicle, from a single tumor line (HT-1376) at a single dose	⚪🟠⚪
	Vs Padcev, low-Nectin-4 model	– 75% TGI vs 53% for Padcev 3 mg/kg x3 at Day 38, with a single radioligand dose benchmarked against a multi-dose ADC schedule and no independent replication	⚪🟠⚪
Human Data (Imaging / Tracer Study, n=20)	**Tumor delivery confirmed**	– 15/15 evaluable patients showed uptake across 5 tumor types including brain metastasis, with rapid blood and tissue clearance	🟢⚪⚪
	Dosimetry projection	– Six cycles of ~8 MBq projected to keep organ doses within safe limits based on full renal profiles from 8 patients	⚪🟠⚪
	Therapeutic-dose safety	– Starting dose cleared with no DLT but no per-patient AE data are public, as tracer-dose tolerability is not informative for therapeutic alpha dose	⚪⚪🔴
Trial Design Flags	**Part 1 primary endpoints**	– Safety (DLT rate and AEs by severity) are the Part 1 primary endpoints, with ORR primary only in Part 2 and no results posted as of Jun 2026	⚪⚪🔴
	PET companion Dx (Part 2 enrollment)	– [64Cu]Cu-AKY-1189 PET required for Part-2 enrollment with no validated IHC cutoff or companion-Dx regulatory precedent	⚪🟠⚪

🟢 No major issues 🟠 Potential risks, addressable 🔴 Major issues / no mitigation

Sources: NCT07020117 — NECTINIUM-2 Phase 1b (Aktis Oncology) • BIO/Citeline Clinical Development Success Rates 2011–2020 • Wong et al., Biostatistics 2019 — phase-transition probabilities

Key Data Gaps

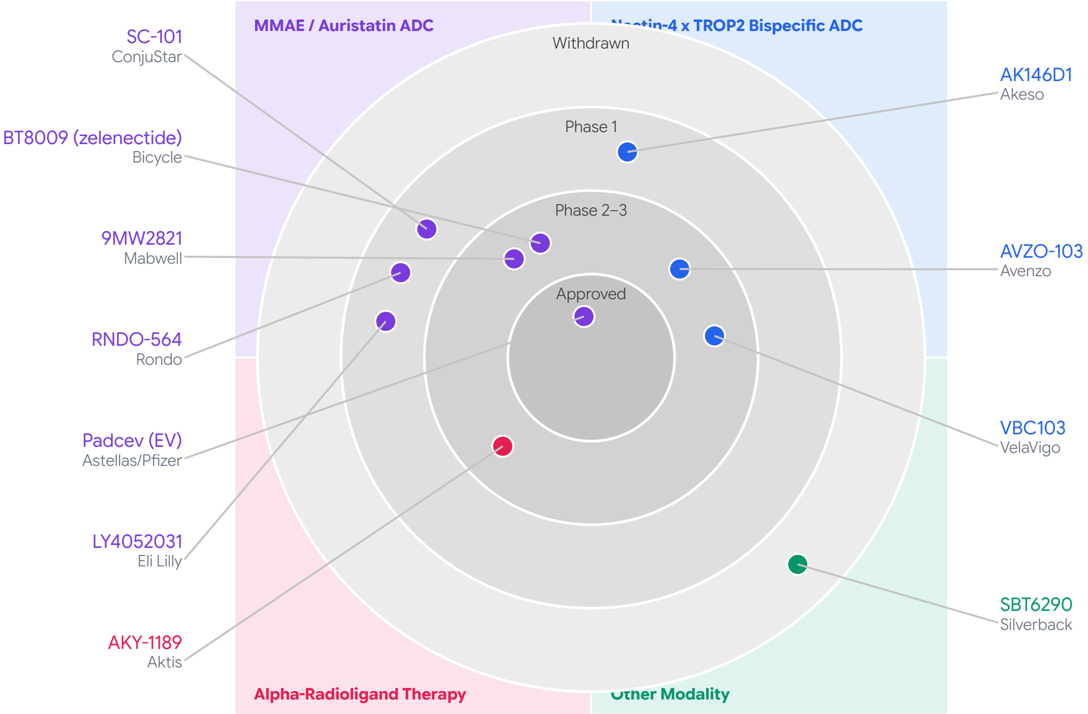
no human efficacy data; tox package incomplete; next gate ≥Q1 2027

	Assessment	Observations	Rating
Clinical Evidence	**Human anti-tumor efficacy (ORR/DoR/PFS)**	– Part-2 ORR not yet read out	○ ○ 🔴
	Human MTD / RP2D	– Dose escalation ongoing, efficacious dose not yet confirmed	○ 🟠 ○
	Dose-response for efficacy	– No human data, non-monotonicity risk unquantified	○ 🟠 ○
Safety & Toxicology	**Marrow & renal safety at therapeutic dose**	– No public repeat-dose toxicology results	○ ○ 🔴
	Genotoxicity / mutagenicity battery	– Not publicly disclosed	○ 🟠 ○
	High-dose CNS / seizure liability	– Not publicly disclosed	○ 🟠 ○
Regulatory & Biomarker	**Registrational endpoint / pivotal design**	– Fast Track granted Feb 2026, no agreed endpoint in public domain	○ 🟠 ○
	PET SUV threshold / post-EV Nectin-4 expression	– Not established, biomarker selection thesis unvalidated	○ ○ 🔴

🟢 No major issues 🟠 Potential risks, addressable 🔴 Major issues / no mitigation

Sources: NCT07020117 — NECTINIUM-2, Phase 1b (ClinicalTrials.gov) • Wong et al. *Biostatistics* 2019 — oncology phase-transition benchmarks

MMAE-saturated Nectin-4 field; AKY-1189 occupies uncontested alpha-RLT lane



Sources: EV-301 — NCT03474107 (ClinicalTrials.gov) • EV-302/KEYNOTE-A39 — NCT04223856 (ClinicalTrials.gov) • LY4052031 NEXUS-01 — NCT06465069 (ClinicalTrials.gov) • NCT07020117 — AKY-1189 NECTINIUM-2 (ClinicalTrials.gov)

Q1 2027 Part 1 readout gates rNPV; supply and regulatory risks cap position sizing

	Assessment	Observations	Rating
Value Drivers	**Target validation**	– EV-301/302 OS HR 0.70 and 0.47 provide clinical and commercial precedent for Nectin-4	● ○ ○
	Alpha-RLT mechanism	– Efflux-independent kill cannot be replicated by MMAE or Topo-I competitors	● ○ ○
	PET companion diagnostic	– [64Cu]Cu-AKY-1189 adds franchise value if validated versus IHC in NECTINIUM-2	○ ● ○
Value Risks	**No human efficacy data**	– Pre-PoC ceiling applies with value re-rating contingent on Q1 2027 Part 1 outcome	○ ○ ●
	Basket class failures	– EV-202 Nectin-4 misses in TNBC (19%) and NSCLC (11%) weaken expansion thesis beyond payload	○ ○ ●
	Competitor compression	– Zelenectide (65% ORR) and LY4052031 (33–47% ORR) arrive 3–4 years before AKY-1189	○ ○ ●
	Ac-225 supply chain	– No confirmed supply agreement with BMS/RayzeBio Phase 3 pause as direct precedent	○ ○ ●
	Financing runway	– $538.5M cash to 2029 removes dilution risk but eliminates seller urgency at a ~$1.7B cap	○ ● ○

● No major issues ● Potential risks, addressable ● Major issues / no mitigation

Sources: NCT07020117 — NECTINIUM-2, Aktis Oncology Phase 1b (ClinicalTrials.gov) • Wong et al. 2019, Biostatistics — oncology phase-transition rates

Four signals in 12 months define the rNPV range before Part 1 data

Near-term value inflections to monitor before the Phase 1b readout:

Lilly lock-up expiry (est. Jul 8, 2026): 13F/Form 4 filings reveal Lilly's strategic intent and cap-table direction

9MW2821 cervical Phase 3 interim (H2 2026): First Phase 3 Nectin-4 read outside urothelial — hit expands basket rNPV, miss signals class failure

EV-302 42.8-month OS update (H2 2026): Raises post-EV survival bar AKY-1189 must clear in refractory mUC

NECTINIUM-2 Part 1 (Q1 2027, NCT07020117): Primary value gate — no DLT-capped dose, renal dose <~23 Gy, ≥1 confirmed RECIST PR in post-EV mUC

Still unresolved before Part 1 data:

Supply (Ac-225 agreement, daughter dosimetry) and deal rights (Lilly ROFN/ROFO, BMS/Novartis/MRL thresholds) unconfirmed

FDA alignment on AA pivotal design and post-AA OS confirmatory requirement unconfirmed

Take-home: At ~$1.7B cap, H2 2026 signals sharpen downside range; Q1 2027 Part 1 data determines whether rNPV justifies current pricing or a re-rate.

Sources: NCT07020117 — NECTINIUM-2, Aktis Oncology Phase 1b (ClinicalTrials.gov) • Wong et al. 2019, Biostatistics — oncology phase-transition rates

Key sources for this assessment

Asset Snapshot — NECTINIUM-2 (NCT07020117) · Padcev EV-302 data and commercial update — pharmaphorum · ACS Cancer Facts & Figures 2024 · Pluvicto $42,500/dose WAC — Novartis press release · BIO Clinical Development Success Rates 2011–2020 · Padcev + Keytruda cost analysis — Prime Therapeutics

Pricing Benchmark — Lutathera buy-and-bill pricing, HCPCS A9513 — $55,000–$61,005/dose · Xofigo FDA prescribing information, NDA 203971 · BIO / Citeline "Clinical Development Success Rates 2011–2020" (2021) · Wong et al. "Estimation of Clinical Trial Success Rates" — Biostatistics 2019 · Novartis FY2024 results — Pluvicto $1.39B (+42%) commercial context

Standard of Care — EV-302/KEYNOTE-A39 (NCT04223856) · JAVELIN Bladder 100 (NCT02603432) · KEYNOTE-045 (NCT02256436) · EV-301 (NCT03474107) · Lutathera FDA prescribing information (accessdata.fda.gov)

Target & Modality Risks — ERA-223 trial — Smith et al., Lancet Oncol 2019 · Padcev (enfortumab vedotin) prescribing information (DailyMed)

Key sources (continued)

Key Data Gaps — [LY4052031 NEXUS-01 — NCT06465069 (ClinicalTrials.gov)]

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